1 Lecture 15: April 14, 2026

Lecture Overview:

1.1 Uniform Convergence

Section Overview:

Recall from last time, we defined **uniform convergence** in terms of a sequence of functions to converge uniformly.

Definition 1.1. Let $\{f_n\}$ be a sequence of functions defined on E. We say $\{f_n\}$ **converges uniformly** on E to a function f if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n > N$,

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in E,$$

or equivalently,

$$\sup_{x \in E} |f_n(x) - f(x)| \leq \varepsilon.$$

Definition 1.2. Let X be a metric space. Let $\mathscr{C}(X)$ denote the set of complex (or real) continuous bounded functions with domain X.

We can define a norm on $\mathscr{C}(X)$ by

$$\|f\| = \sup_{x \in X} |f(x)|,$$

which satisfies $\|f\| \geq 0$ with equality if and only if $f = 0$, and the triangle inequality. We verify all three norm conditions:

Proof. **(1) Nonnegativity.** Since $|f(x)| \geq 0$ for all $x \in X$, we have $\|f\| = \sup_{x \in X} |f(x)| \geq 0$. If $f = 0$, then $|f(x)| = 0$ for all x, so $\|f\| = 0$. Conversely, if $\|f\| = 0$, then $|f(x)| \leq 0$ for all $x \in X$, so $f(x) = 0$ for all x, i.e., $f = 0$.

(2) Scalar homogeneity. For any scalar c,

$$\|cf\| = \sup_{x \in X} |cf(x)| = \sup_{x \in X} |c||f(x)| = |c| \sup_{x \in X} |f(x)| = |c| \, \|f\|.$$

(3) Triangle inequality. For any $f, g \in \mathscr{C}(X)$ and $x \in X$,

$$|f(x) + g(x)| \leq |f(x)| + |g(x)| \leq \|f\| + \|g\|.$$

Taking the supremum over $x \in X$ gives $\|f + g\| \leq \|f\| + \|g\|$. □

Define a metric on $\mathscr{C}(X)$ by

$$d(f, g) = \|f - g\|.$$

Theorem 1.3. *If $f_n \to f$ uniformly on E, then for all $x \in E'$, if $\lim_{t \to x} f_n(t)$ exists ...*

Theorem 1.4. *If f_n is continuous for all n, then*

$$f(x) = \lim_{n \to \infty} f_n(x) = \lim_{t \to x} f(t).$$

Theorem 1.5. *$\mathscr{C}(X)$ is a complete metric space.*

Recall that **complete** means that every Cauchy sequence converges.

Proof. Let $\{f_n\}$ be Cauchy in $\mathscr{C}(X)$. Let $f(x) = \lim_{n\to\infty} f_n(x)$. We will show $f \in \mathscr{C}(X)$.

(1) f is continuous. Since $\{f_n\}$ is Cauchy, for every $\varepsilon > 0$ there exists N such that for $n, m > N$, $\|f_n - f_m\| < \varepsilon$, i.e., $|f_n(x) - f_m(x)| < \varepsilon$ for all $x \in X$. Letting $m \to \infty$ gives $|f_n(x) - f(x)| \le \varepsilon$ for all $x \in X$ and $n > N$, so $f_n \to f$ uniformly on X. Since each f_n is continuous, by the theorem above f is continuous.

(2) f is bounded. Fix $\varepsilon = 1$. By the uniform convergence above, there exists N such that $\|f_N - f\| \le 1$, i.e., $|f_N(x) - f(x)| \le 1$ for all $x \in X$. Since $f_N \in \mathscr{C}(X)$, f_N is bounded, so there exists M with $|f_N(x)| \le M$ for all x. Then

$$|f(x)| \le |f(x) - f_N(x)| + |f_N(x)| \le 1 + M$$

for all $x \in X$, so f is bounded. Thus $f \in \mathscr{C}(X)$, and $f_n \to f$ in the metric d, proving $\mathscr{C}(X)$ is complete. $\square$

1.2 Uniform Convergence and Integration

Section Overview:

Theorem 1.6. *Let α be monotonically increasing on $[a,b]$, and let $\mathscr{R}(\alpha)$ denote the set of Riemann–Stieltjes integrable functions on $[a,b]$ with respect to α. Suppose $f_n \in \mathscr{R}(\alpha)$ for all n and $f_n \to f$ uniformly on $[a,b]$. Then $f \in \mathscr{R}(\alpha)$ and*

$$\int_a^b f\,d\alpha = \lim_{n\to\infty} \int_a^b f_n\,d\alpha.$$

Proof. Let $\varepsilon_n = \sup_{x\in[a,b]} |f_n(x) - f(x)|$. Since $f_n \to f$ uniformly, $\varepsilon_n \to 0$. For all $x \in [a,b]$,

$$f_n(x) - \varepsilon_n \le f(x) \le f_n(x) + \varepsilon_n.$$

Since $f_n \in \mathscr{R}(\alpha)$, integrating gives

$$\int_a^b (f_n - \varepsilon_n)\,d\alpha \le \underline{\int_a^b} f\,d\alpha \le \overline{\int_a^b} f\,d\alpha \le \int_a^b (f_n + \varepsilon_n)\,d\alpha,$$

so

$$0 \le \overline{\int_a^b} f\,d\alpha - \underline{\int_a^b} f\,d\alpha \le 2\varepsilon_n[\alpha(b) - \alpha(a)].$$

Since $\varepsilon_n \to 0$, the upper and lower integrals of f agree, so $f \in \mathscr{R}(\alpha)$. Moreover,

$$\left|\int_a^b f\,d\alpha - \int_a^b f_n\,d\alpha\right| = \left|\int_a^b (f - f_n)\,d\alpha\right| \le \varepsilon_n[\alpha(b) - \alpha(a)] \to 0,$$

so $\int_a^b f\,d\alpha = \lim_{n\to\infty} \int_a^b f_n\,d\alpha$. $\square$